FIITJEE

JEE(Advanced)-2018

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS
CONCEPT RECAPITULATION TEST – III
PAPER-2

Q. No.	PHYSICS	Q. No.	CHEMISTRY	Q. No.	MATHEMATICS
1.	C	24.	A	47.	D
2.	B	25.	D	48.	A
3.	B	26.	B	49.	A
4.	AD	27.	BCD	50.	ACD
5.	BD	28.	BC	51.	ABC
6.	AB	29.	AD	52.	AC
7.	AC	30.	ABC	53.	BC
8.	ABC	31.	ABC	54.	BC
9.	C	32.	C	55.	C
10.	B	33.	D	56.	B
11.	2	34.	8	57.	7
12.	5	35.	9	58.	2
13.	2	36.	6	59.	0
14.	9	37.	9	60.	1
15.	2	38.	7	61.	5
16.	2	39.	6	62.	6
17.	8	40.	7	63.	7
18.	8	41.	4	64.	9
19.	5	42.	3	65.	0
20.	8	43.	6	66.	8
21.	00772.00	44.	00007.00	67.	00048.00
22.	01057.50	45.	00044.00	68.	00012.00
23.	00200.00	46.	00000.00	69.	00036.00

Physics

PART – I

SECTION – A

1. C

$W = P_0\Delta V = P_0V_0$

and $\Delta U = 2P_0\left(2V_0 - V_0\right) = 2P_0V_0$

$\therefore Q = W + \Delta U = 3P_0V_0$

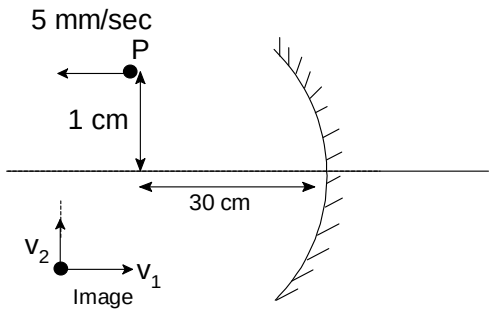
2. B

$T = Ax\rho g\left(x = \text{distance from lower end}\right)$

$\text{Stress} = \dfrac{T}{A} = x\rho g$

$\Delta\ell = \dfrac{\rho g\ell^2}{2Y}$

3. B



$\dfrac{1}{v} + \dfrac{1}{-30} = \dfrac{1}{-20}$

$v = -60$

$m = \dfrac{y_i}{y_0} = -\dfrac{v}{u}$

$y_i = -2\text{cm}$

for $\vec{v}_1 \quad \vec{v}_1 = -\dfrac{v^2}{u^2}\left(\vec{v}_P\right)$

$= -4\left(-5\right) = 20 \text{ mm / sec}$

for $\vec{v}_2 \Rightarrow \dfrac{y_i}{y_0} = -\dfrac{v}{u}$

$y_iu = -y_0v$

on differentiating the equation with respect to time

$\dfrac{dy_i}{dt}(u) + y_i\dfrac{du}{dt} = y_0\dfrac{dv}{dt} \qquad \left(\dfrac{dy_0}{dt} = 0\right)$

$\dfrac{dy_i}{dt}\left(-30\right) + \left(-2\right)\left(-5\right) = -\left(20\right)$

$\dfrac{dy_i}{dt} = 1\,\text{mm / sec}$

$v_i = 20\hat{i} + \hat{j} \text{ mm / sec}$

4. AD

(a) We can easily see that charge q is placed symmetrically to surface ABCD, ABSR and ADQR. Charge q is also placed symmetrically to rest of the surfaces.
If the flux through the surface ABCD is x and through RSPQ is y
Now by Gauss law

$$\frac{q_{in}}{\varepsilon_0} = \phi$$

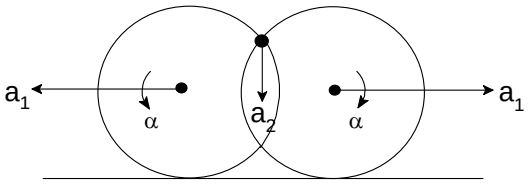
$$\Rightarrow 3x + 3y = \frac{q}{\varepsilon_0}$$

$$\Rightarrow x + y = \frac{q}{3\varepsilon_0}$$

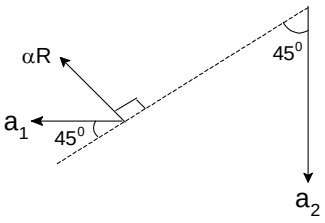
(d) Flux through two surfaces are not same flux via ABCD is larger.

Ans: (a) $\frac{q}{3\varepsilon_0}$ (b) Flux through two surfaces are not same flux via ABCD is larger.

5. BD

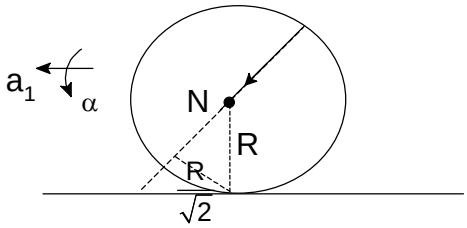


Constraint relation



$$a_1 \cos 45^0 = a_2 \cos 45^0$$

$$\Rightarrow a_1 = a_2 = a$$



$$\frac{NR}{\sqrt{2}} = 2mR^2\left(\frac{a}{R}\right)$$

$$N = 2\sqrt{2}ma \quad .....(1)$$

$$mg - \sqrt{2}N = ma \quad .....(2)$$

from $(1)\&(2)$

we get $a = \frac{g}{5}$

6. AB

$$E_{ind} = L\frac{di}{dt} = Bv\ell \Rightarrow di = \frac{B\ell}{L}, (vdt) = \frac{B\ell}{L}.dx$$

$\therefore i = \frac{B\ell}{L}.x$

$F_{mag} = -Bi\ell = -\frac{B^2\ell^2}{L}.X\left(\longleftarrow\right)\left(\therefore \text{SHM of rod}\right)$

$\omega = \sqrt{\frac{B^2\ell^2}{mL}}$

using SHM equation $v^2 = v_{max}^2 - \omega^2 x^2$

$\Rightarrow x^2 = \frac{3v_{max}^2}{4\omega^2}$

$x = \sqrt{\frac{3J^2L}{4mB^2\ell^2}}$

$\therefore i = \frac{B\ell}{L}.x = \sqrt{\frac{3J^2}{4mL}}$

7. $N - 2mg = 2ma_1$ .....(i)

$N + mg - T = ma_2$ .....(ii)

$T - 3mg = 3ma_3$ .....(iii)

$a_1 = 4$ and $a_2 = a_3 \Rightarrow$ $N = 2mg + 2m$.....(iv)

$\Rightarrow a_2 = 2\ m/s^2$ (by equations)

$\Rightarrow a_{rel} = 2m/s^2 \Rightarrow S_{rel} = \frac{1}{2}a_{rel}t^2 = \frac{1}{2}\times 2\times 1 = 1m.$

$a_3 = 4;\ a_2 = -a_1$

$\Rightarrow T = 3mg + 12m$

by(ii) $N + mg - 3mg - 12m = -ma_1$

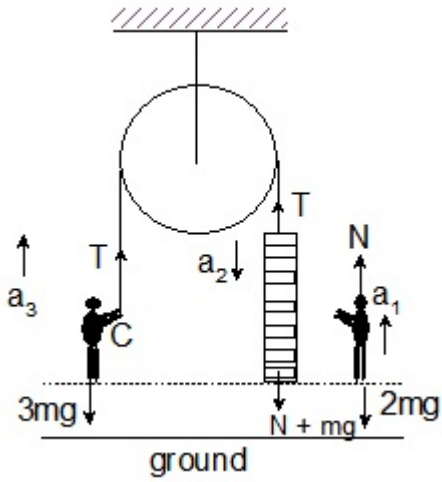
$\Rightarrow N = 2mg + 12m - ma_1$

by(i) $2mg + 12m - ma_1 - 2mg = 2ma_1 \Rightarrow a_1 = 4$

$\Rightarrow a_{rel} = 0 \Rightarrow S_{rel} = 0$

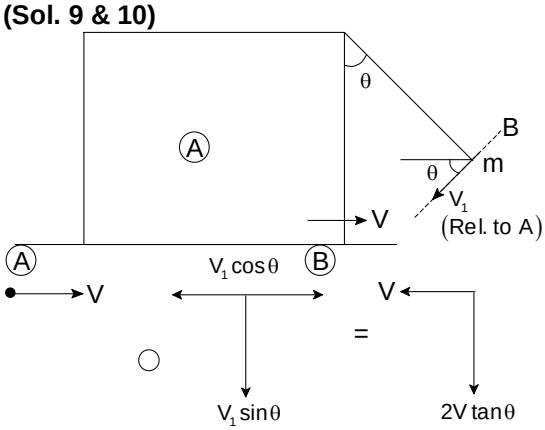
$a_3 = 4; a_1 = 4 \Rightarrow T = 3mg + 12m = ma_2 \Rightarrow 42$

$\Rightarrow a_2 = 4m/s^2$ upward



8. ABC

(Sol. 9 & 10)



Applying conservation of linear momentum.

$$mV = m(v_1\cos\theta - v) \Rightarrow V_1 = \frac{2V}{\cos\theta}\left[V_1 = \frac{2V}{\cos\theta}\right]$$

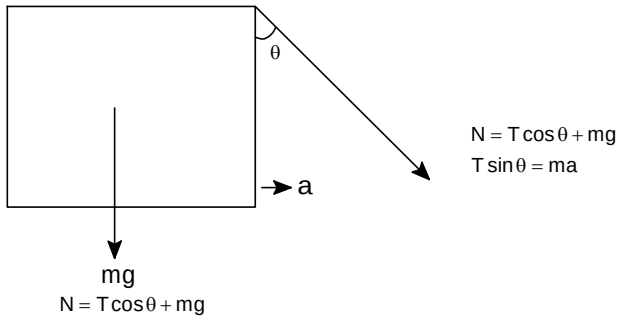
Applying conservation of mechanical energy

$$mg\ell\cos\theta = \frac{1}{2}mV^2 + \frac{1}{2}m\left(V^2 + 4V^2\tan^2\theta\right)$$

$$= \frac{1}{2}mV^2\left(2 + 4\tan^2\theta\right) = mV^2\left(1 + 2\tan^2\theta\right)$$

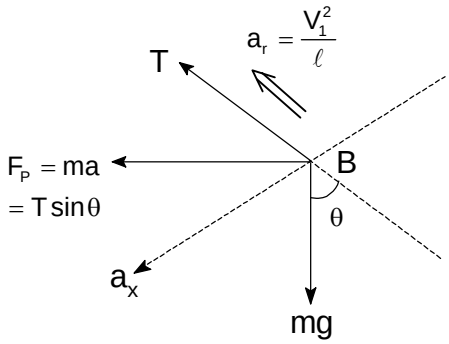
$$V = \sqrt{\frac{g\ell\cos\theta}{1 + 2\tan^2\theta}}$$

$$V_1 = 2\sqrt{\frac{g\ell}{\cos\theta\left(1 + 2\tan^2\theta\right)}} = 2\sqrt{\frac{g\ell\cos\theta}{1 + \sin^2\theta}}$$



$T\sin\theta = ma$

Motion B w.r.t. A



$$T + T\sin^2\theta - mg\cos\theta = \frac{mV_1^2}{\ell}$$

$$T\left(1 + \sin^2\theta\right) = mg\cos\theta + \frac{m}{\ell}\left(4.\frac{g\ell\cos\theta}{1 + \sin^2\theta}\right)$$

$$T\left(1 + \sin^2\theta\right) = mg\left(\cos\theta + \frac{4\cos\theta}{1 + \sin^2\theta}\right)$$

$$T\left(1 + \sin^2\theta\right) = mg\cos\theta\left(\frac{5 + \sin^2\theta}{1 + \sin^2\theta}\right)$$

$$T = mg\cos\theta\left(\frac{5 + \sin^2\theta}{\left(1 + \sin^2\theta\right)^2}\right)$$

$$N = T\cos\theta + mg = mg\left(1 + \frac{\cos^2\theta\left(5 + \sin^2\theta\right)}{\left(1 + \sin^2\theta\right)^2}\right)$$

SECTION – C

11. $a_{max} = \omega^2 A_{min} = g$

$$\omega = 2\pi f = \frac{2\pi v}{\lambda}$$

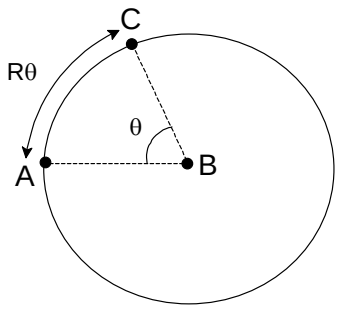
$$v = \sqrt{\frac{F}{\mu}}$$

$$A_{min} = \frac{g}{\omega^2}$$

$$= g \times \frac{\mu\lambda^2}{4\pi^2 F} = 2 \times 10^{-3} m = 2mm$$

12. Energy required to just remove the electron = 13.6 eV
∴ Energy required = 13.6 + 16.4 =30 eV
If E be the photon energy 25%
E = 30 eV, E = 120 eV = 24×5 eV
X = 5

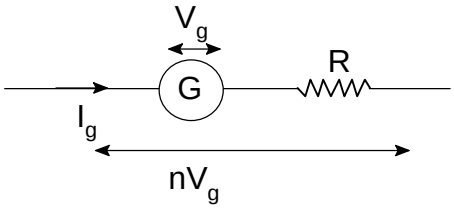
13. Let they meet at C



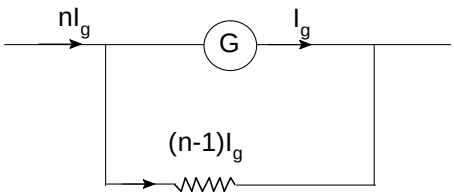
$$\frac{R}{v}=\sqrt{\frac{2R\theta}{a}}$$

$$v=R\sqrt{\frac{a}{2R\theta}}=\sqrt{\frac{aR}{2\theta}}$$

14.



$$R=\frac{(n-1)V_g}{I_g}=(n-1)\times G$$



$$S=\frac{G}{n-1};\ G=\sqrt{RS}$$

15. Since the angle of E with horizontal and vertical is 45^0, magnitude of Eq and Eq' must be equal.

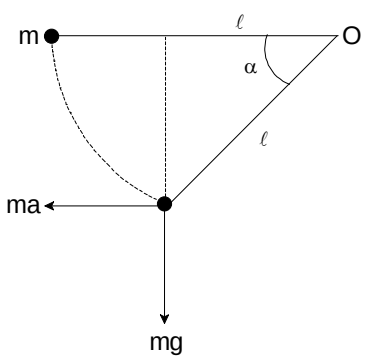
$$Eq'=k\frac{|q'|}{(2d)^2}=k\frac{|q'|}{4d^2}\qquad Eq=k\frac{|q|}{d^2}$$

$$q'=-4q$$

$$\text{potential of point A } V_A=V'+V=\frac{-k4q}{2d}+k\frac{q}{d}=-k\frac{q}{d}$$

$$=\frac{-q}{4\pi\varepsilon_0 d}$$

16. Applying W.E. T with respect to O.



$$mg\ell\sin\alpha=ma\ell(1-\cos\alpha)$$

$$\frac{2\sin\frac{\alpha}{2}\cos\frac{\alpha}{2}}{2\sin^2\frac{\alpha}{2}}=\frac{a}{g};\ \cot\frac{\alpha}{2}=\frac{4}{3};\ \tan\frac{\alpha}{2}=\frac{3}{4}$$

the maximum vertical displacement

$$=\ell\sin\alpha=\frac{24\ell}{25}=\frac{48}{25};\ \ell=2m$$

17. For fluid in tube $\sum F = m\,a_{cm}$

$$(P - P_0)A = A\ell\rho\omega^2\left(L - \frac{\ell}{2}\right)$$

$$P = P_0 + \ell\rho\omega^2\left(L - \frac{\ell}{2}\right)$$

By applying Bernouli theorem at end of the tube

$$P_0 + \ell\rho\omega^2\left(L - \frac{\ell}{2}\right) = P_0 + \frac{1}{2}\rho v^2$$

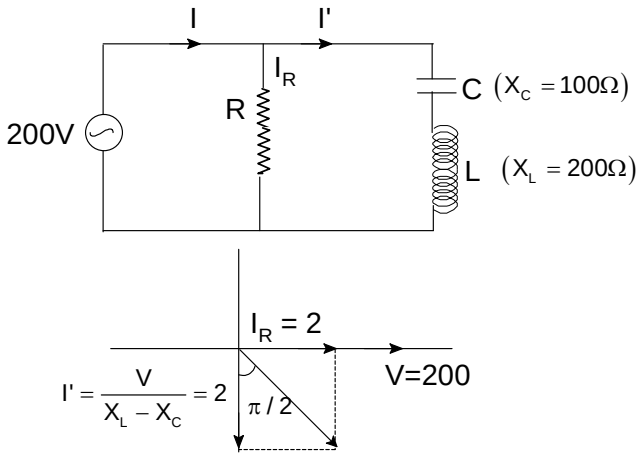
$$v = \sqrt{\frac{2\omega^2\ell^2}{2}\left(\frac{2L}{\ell} - 1\right)} = \omega\ell\sqrt{\frac{2L}{\ell} - 1}$$

$$= 10 \times 0.4\sqrt{\frac{2\times 1}{0.4} - 1} = 8\,m/sec$$

18. $I_r = \frac{V}{R} = \frac{200}{100} = 2A$

$$I' = \frac{V}{X_L - X_C} = \frac{200}{100} = 2A$$

$$I = \sqrt{i_R^2 + I'^2} = 2\sqrt{2}\ \text{Amp.}$$



19. The F.B.D. of wire PQ is

The force due to surface tension = $F_{ST} = 2T \times 2\,AD\,\tan\theta$

$F_{sT} = (2T)2AD\tan\theta$ $\qquad$ $F_{ST} = (2T)2(AD + x)\tan\theta$

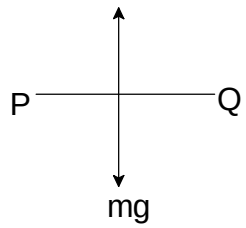


Figure (a)

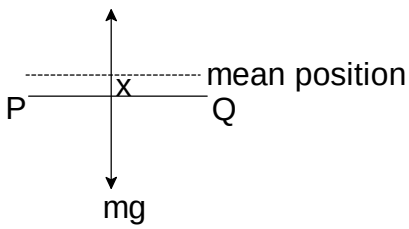


Figure (b)

For wire to be in equilibrium (Figure (a))

$4T\ AD\ \tan\theta = mg$ .....(i)

If the wire PQ is at a distance x below the mean position, the restoring force on the wire is (Figure (b))

$-ma = 4T\tan\theta(AD + x) - mg = 4T\tan\theta x$

Hence the wire PQ executives SHM

$$a = -\frac{4T}{m}\tan\theta x$$

comparing with $a = -\omega^2 x$ we get

$$\omega^2 = \frac{4T}{m}\tan\theta$$

or $T = 2\pi\sqrt{\frac{m}{4T\tan\theta}} = 2\pi\sqrt{\frac{1\times10^{-3}}{4\times25\times10^{-3}}} = \frac{\pi}{5}s$

20. $\frac{4}{3}\pi R^3.\rho_2.g + Ah.\rho_1 g = \frac{4}{3}\pi R^3.\rho_W.g + Ah_1.\rho_w g$

$$\rightarrow = \left[\frac{3A\left(h_1\rho_w - h\rho_1\right)}{4\pi\left(\rho_2 - \rho_w\right)}\right]^{1/3}$$

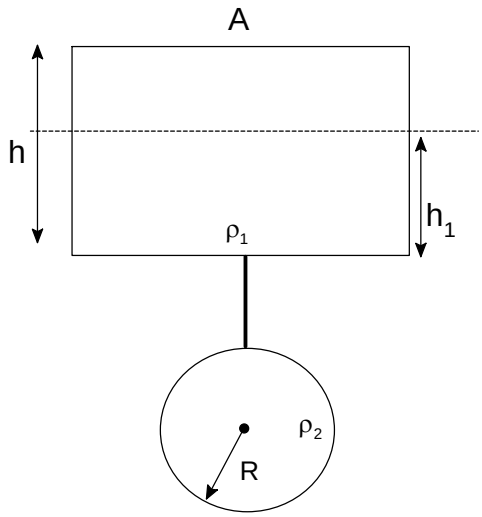
Using values

A = 11 cm^2; h_1 = 4 cm; $\rho_w = 1\,gm/cm^3$;

$\rho_1 = 0.5\ gm/cm^3$; $\rho_2 = 8gm/cm^3$

$$R = \left[\frac{3\times11\left(4\times1-6\times0.5\right)}{4\times\left(\frac{22}{7}\right)\times\left(8-1\right)}\right] = \left(\frac{3}{8}\right)^{1/3} cm$$

$$\Rightarrow R^3 = \frac{3}{8}$$



SECTION – D

21. Rate of formation of nuclei at time t is

$\frac{dN}{dt} = \alpha - \lambda N$, where N = number of nuclei at that time.

$$N = \frac{\alpha}{\lambda}\left(1-e^{-\lambda t}\right)$$

Number of nuclei disintegrated in time t, $\left[\alpha t - \frac{\alpha}{\lambda}\left(1-e^{-\lambda t}\right)\right]$

$\therefore$ Energy released till time t, $E = E_0 N_d = E_0\left[\alpha t - \frac{\alpha}{\lambda}\left(1-e^{-\lambda t}\right)\right]$

20% of this energy is used for heating water,

$$0.2E_0\left[\alpha t - \frac{\alpha}{\lambda}\left(1-e^{-\lambda t}\right)\right] = ms\Delta\theta$$

$$\Delta\theta = \frac{0.2E_0\left[\alpha t - \frac{\alpha}{\lambda}\left(1-e^{-\lambda t}\right)\right]}{ms}$$

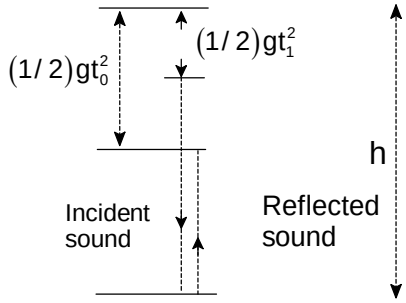
At $t = T_{1/2}$

$$\Delta\theta = \frac{0.2E_0\left[2\lambda T_{1/2} - \frac{2\lambda}{\lambda}\left(1-\frac{1}{2}\right)\right]}{ms}$$

$$\Delta\theta = \frac{0.2E_0\left[0.386\right]}{ms} = 772K$$

22. Let the sound observed by the parachutist at t_0 = 12s be produced at t_1s. velocity of source at the instant of emitting sound = gt_1 and velocity of observer at the instant of observing same sound = gt_0. Hence the relation between apparent frequency f' and original frequency f will be

$$f' = f\left(\frac{v + gt_0}{v - gt_1}\right)$$



Here $f = 800\text{Hz}$, $g = 10\text{m/s}^2$, $v = 330\text{ m/s}$,

$t_0 = 12\text{s}$ and $f' = 800 + 700 = 1500\text{Hz}$

Putting these, we get

$t_1 = 9\text{s}$

Now the distance traveled by sound in $(t_0 - t_1)$sec

is $v(t_0 - t_1) = \left(h - \frac{1}{2}gt_0^2\right) + \left(h - \frac{1}{2}gt_1^2\right)$

Putting the values, we get

$h = 1057.5\text{ m}$

23. $2mg - A x \rho g - T = 2ma \quad ....(i)$

$T - mg = ma \quad ...(ii)$

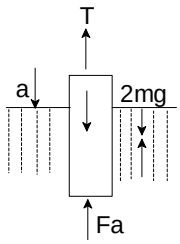
From (i) and (ii) $\Rightarrow mg - A x \rho g = 3ma \Rightarrow a = \frac{g}{3} - \frac{A\rho x}{3m}g = k_1 - k_2 x$

$v\frac{dv}{dx} = k_1 - k_2 x$, where $k_1 = \frac{g}{3}$ and $k_2 = \frac{\rho g A}{3m}$

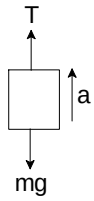
$\Rightarrow \int_0^0 v\,dv = \int_0^L (k_1 - k_2 x)\,dx$

$\Rightarrow 0 = k_1 L - \frac{k_2 L^2}{2}$

$\Rightarrow \frac{g}{3} = \frac{A\rho g}{3m}.\frac{L}{2} \Rightarrow \rho = \frac{2m}{AL}$



F.B.D. of 2m



F.B.D of m

Chemistry

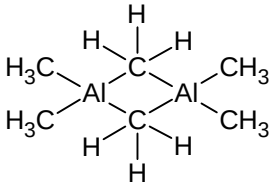
PART – II

SECTION – A

24. Correct balanced equation is:

$$2MnO_4^- + 5H_2O_2 + 6H^+ \longrightarrow 2Mn^{+2} + 5O_2 + 8H_2O$$

25. A, B, C are correct statements



has banana bond

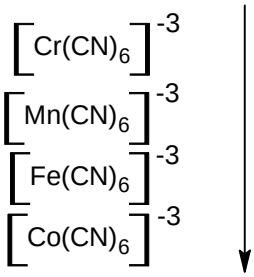
26. $CH_3CH_2\overset{O}{\overset{\|}{C}}H < \underset{Cl}{\overset{|}{C}}H_2CH_2\overset{O}{\overset{\|}{C}}H < CH_3\underset{Cl}{\overset{|}{C}}H\overset{O}{\overset{\|}{C}}H < CH_3CF_2\overset{O}{\overset{\|}{C}}H$

Smallest K — Largest K

27. In acidic medium nucleophilicity of enol decide the rate of reaction, but in basic medium stability of enolate decide the rate of reaction.

28. (A) not reducing: an acetal ending in "oside"
(B) reducing: a hemiacetal ending in ose
(C) reducing: one of ring has a hemiacetal
(D) not reducing: all anomeric carbons are in acetal form

29.



$[Cr(CN)_6]^{-3}$
$[Mn(CN)_6]^{-3}$
$[Fe(CN)_6]^{-3}$
$[Co(CN)_6]^{-3}$

Synergic bonding

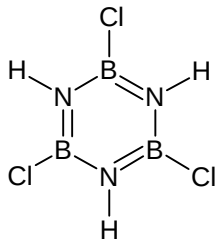
due to more e^- in d orbital

30. A (after combination) $\Rightarrow$ 0.5 M NH_3 + 0.5 M CH_3COOH
B (after combination) $\Rightarrow$ 0.5 M HCO_3^- + 0.5 M CO_3^{-2}
C (after combination) $\Rightarrow$ 0.5 M CH_3COO^- + 0.5 M CH_3COOH

31. A $\rightarrow$ NGP
B $\rightarrow$ NGP
C $\rightarrow$ NGP

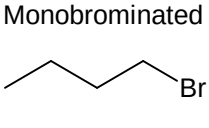
32. $B_3N_3H_6$ is called borazine. Borazon is the cubic form of $(BN)_x$

33. Structure of $B_3N_3Cl_3H_3$

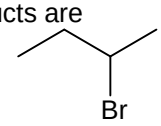


SECTION – C

34. Monobrominated products are

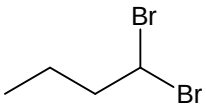


(1)

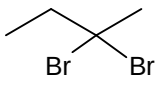


(d/l)
(2)

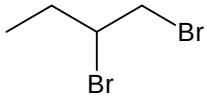
Dibrominated products are



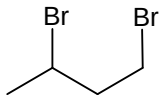
(1)



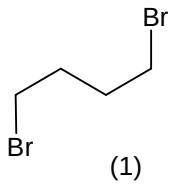
(1)



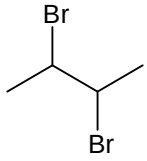
(d/l)
(2)



(d/l)
(2)



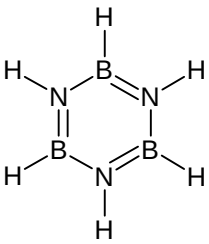
(1)



(d/l/Meso)
(3)

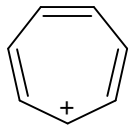
All d/l pairs are optically active.

35.

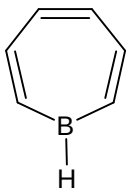


Borazine(aromatic)

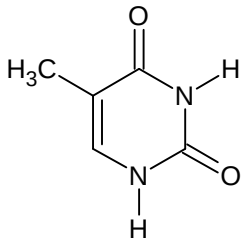
Graphite (aromatic)



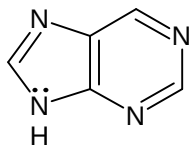
(aromatic)



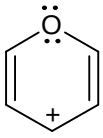
(aromatic)



(aromatic)



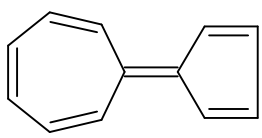
(aromatic)



(aromatic)



(aromatic)



(aromatic)

36. A, B, C, D, F, G show mutarotation because these molecules have OH group on anomeric (at least one) carbon.

37. all are more basic than NH_3 (in water)

38. A, B, C, D, E, F, G, I are optically active H is optically inactive (it has plane of symmetry)

39. $Ni^{+2} \rightarrow a$
$Ni^{+3} \rightarrow 0.98 - a$
then
$a \times (+2) + (0.98 - a)(+3) - 2 \times 1.00 = 0$
$2a + 2.94 - 3a - 2 = 0$
$-a - 0.94 = 0$
$a = 0.94$
% of $Ni^{+2} = \frac{0.94}{0.98} \times 100 \approx 96$
% of $Ni^{+3} \approx 4$
Now $\frac{x}{4y} = \frac{96}{4 \times 4} = 6$

40. $\Delta T = \frac{1000 \times K_b \times w}{m \times W}$
$0.01 = \frac{1000 \times 0.5 \times w}{180 \times 100}$
$w = 0.36\,g$
$\therefore$ no of molecules of glucose in 0.36 g = $(6.023 \times 10^{23} \times 0.36)/180$
$= 1.2 \times 10^{21}$ molecule
Hence $\frac{x}{3} = \frac{21}{3} = 7$

41. $NaOH \Rightarrow Na^{+}\ {}^{-}O - H$
$Na_2O_2 \Rightarrow 2Na^{+}\ {}^{-}O - O^{-}$
$BaO_2 \Rightarrow Ba^{+2}\ {}^{-}O - O^{-}$
$Na_2S_2 \Rightarrow 2Na^{+}\ {}^{-}S - S^{-}$

42. yellow ppt: AgBr, $PbCrO_4$, PbI_2

43. I, II, III, IV, V & VII are correct
VI is incorrect, correct order is $Cl_2 > Br_2 > F_2 > I_2$

SECTION – D

44. $C_{v,m} = \frac{\Delta U}{n\Delta T} = \frac{40 \times 8.314\ J}{2 \times 8\ mol\,K} = \frac{5}{2} R$
$C_{P,m} = \frac{7}{2} R$ & $\gamma = \frac{7}{5}$
Reversible adiabatic process

$P_1V_1^{\gamma} = P_2V_2^{\gamma}$

$\frac{P_1}{P_2} = \left(\frac{V_2}{V_1}\right)^{\gamma} = (32)^{\frac{7}{5}} = 2^7$

$\log\left(\frac{P_1}{P_2}\right) = 7\log 2 = x\log 2$

$\therefore x = 7$

45. $\frac{M_{eq}}{M_{initial}} = \left(\frac{1}{1+\alpha}\right)$

$M_{eq} = \frac{60}{1.2} = 50$

$V.D_{eq} = y = 25$

After adding $Ne(g)$

$M_{mix} = \frac{0.6\times 50 + 0.4\times 20}{1}$

$= 30 + 8$

$V.D_{eq} = z = \frac{38}{2} = 19$

$\therefore y + z = 44$

46. (b) 2 radial nodes so can't be of 2s-orbital
(c) Ψ depends on 'θ' so can't be of s orbital
(d) no radial nodes for given Ψ, so can't be of 2s orbital
So only 'a' could be correct and at $r = 2a_0$: $\Psi = 0 \Rightarrow \Psi^2 = 0$ i.e. probability = 0

Mathematics

PART – III

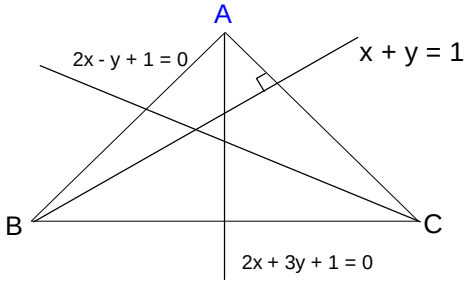
SECTION – A

47. maximum number of trials = 20 + 19 + … + 2 + 1 = $\frac{20\times 21}{2} = 210$

48. Now, Arranging the lines in decreasing order of their slopes, we get
$L_1 = 3x - y + 1 = 0$; $L_2 = x - y + 5 = 0$; $L_3 = 2x + y - 3 = 0$

tanA = $\frac{3-1}{1+3} = \frac{1}{2}$
tanB = -3
tanC = 1
the required cubic equation
(x – 1/2) (x + 3) (x–1) **=** 0
(2x – 1) (x + 3) (x–1) **=** 0

49. Slope of AC = - 1



Let slope of BC = m
Then $\left|\frac{2-1}{1+2}\right| = \left|\frac{m-2}{1+2m}\right| \Rightarrow m = 7$
Let equation of BC is y = 7x + C
So $B = \left(\frac{1-C}{8}, \frac{7+C}{8}\right), C \equiv \left(\frac{1-C}{5}, \frac{7-2C}{5}\right)$
Mid point of BC lie on 2x + 3y + 1 = 0
So $C = \frac{379}{59}$

50. Eg. x + y + z = 1
x + y + z = 2
x + y + z = 3
here $\Delta = \Delta_x = \Delta_y = \Delta_z = 0$ and equation have no solution.

51. $S = \left(n^2 - n + 2\right)2^n - 1$

52. f ′ (x) = 0 $\Rightarrow$ (sin x + cos x) (non zero quantity) = 0
$\Rightarrow$ tan x = – 1 $\Rightarrow$ x = 3π/4 or 7π/4 .
Global Min = x = 2 n π + (3π/4) ; global max = x = 2 n π + (7π/4)
M = $\frac{4}{8-\sqrt{2}}$; m = $\frac{2\sqrt{2}}{4\sqrt{2}+1}$]

53. One root is 2.

54. point of intersection of diagonal which are inside the polygon is ${}^{10}C_4 = 210$
Points on the polygon is 10 but each point is point of intersection of 21 pair of diagonals.
Points outside the polygon = ${}^{{}^{10}C_2}C_2 - 10.{}^{7}C_2 - {}^{10}C_4$
= 175

(Sol. Q. 55-56)

let $\int_0^{\frac{\pi}{2}} \sin tg(t)dt$ = A

So g(x) = (1-2A)cosx
Finally, g(x) = cosx/2

SECTION – C

57. $xg'(x) = g(x) + px^{p-1}$
It's a first order linear differential equation.
$g(x) = \frac{p}{p-2}x^{p-1} - \frac{2}{p-2}x$
So p= 3 or 4

58. p_1 and p_2 are roots of $f'(x) = 0$

59. S = 0

60. normal at (t, t^2) is $2t(y-t^2) = -x+t$.
Passes through $(0, \frac{3}{2})$

61. A one to one function and its inverse are symmetric across the line y = x. Thus x and y intercept are interchanged and the sum is the same i.e. 5.

62. Minimum value of $\frac{r_1 r_2 r_3}{r^3}$ is 9

63. locus is $x^2 + y^2 + z^2 + 5xy + 5yz + 5zx - 4x - 4y - 4z + 2 = 0$

64. ${}^{18}C_3 - {}^{7}C_3 - {}^{7}C_3 - {}^{7}C_3 = 711$

65. constant term = $\Delta(0) = 0$
Coefficient of x = $\Delta'(0) = 0$

66. $\lim_{x\to 0} \frac{\cos x + p^3 \sin 2^3 x - 1}{x} = e^{\frac{5}{2}}$

$\lim_{x\to 0} \frac{p^3 \sin q^3 x - 2\sin^2 \frac{x}{2}}{x} = \frac{5}{2}$

By L′hospital rule
$p^3 q^3 = 2^4 \Rightarrow pq = 8$

SECTION – D

67. $z = -\omega$ or $-\omega^2$, also $(z^3 + \frac{1}{z^3})^2 = 4$ etc

$\Rightarrow$ 32 + 16 = 48

68. Let the altitude of the 1st Δ = h
altitude of the 2nd $\Delta = \sqrt{3}h$
altitude of the 3rd Δ = 3h etc.
now side of the 1st Δ is 'a_1' = $\frac{2h}{\sqrt{3}}$

$\therefore$ $a_2 = 2h$

$a_3 = 2\sqrt{3}\,h$

|||ly $a_6 = 18h$

perimeter = 12 Ans.

69. $\Delta(a) = 4\pi a^2$